Math566: Combinatorial Theory

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Abstract

This is the note containning my personal thoughts as well as lecture notes, course content include some basic algebraic combinatorics. My course instructor is Prof. Shiyue Li.

Contents

This chapter basically collect some linear algebra lemma that maybe helpful across the course.

Lemma 0.0.1. Given A to be a matrix, who has the eigenvalues to be $\lambda_1, \dots, \lambda_p$, then the eigenvalues of the matrix $A + c\,\mathrm{Id}$ where $c \in \mathbb{C}$ is $\lambda_1 + c, \dots, \lambda_p + c$.

Lemma 0.0.2. Let A be as above, the eigenvalues of A^l are $\lambda_1^l, \dots, \lambda_p^l$.

Lemma 0.0.3. Let $f \in \mathbb{C}[t]$, then $f(A)$ has eigenvalues being $\{f(\lambda_i)\}_{i=1}^p$.

Lemma 0.0.4. If A is a real and symmetric matrix, by spectral theorem for Hermitian product, we have $\lambda_i \in \mathbb{R}$.

Lemma 0.0.5. Let $\{\alpha_i\}_{i=1}^r$ and $\{\beta_i\}_{i=1}^s$ be non-zero complex numbers, if for every $l \geq 1$, we have:

$$\sum_{i=1}^r \alpha_i^l = \sum_{i=1}^s \beta_i^l$$

then $r = s$ and $\{\alpha_i\}$ is permutation of $\{\beta_i\}$.

Theorem 0.0.1 (Cauchy-Binet Theorem). Let A, B be $n \times m$ and $m \times n$ matrices, if $n \leq m$, then:

$$\det(AB) = \sum_{|S|=n, S\subseteq[m]} \det(A[S]) \det(B^\intercal[S])$$

If $n > m$, then:

$$\det(AB) = 0$$

Theorem 0.0.2 (Laplacian Expansion Theorem). Let X, Y be matrices of dimension $n \times (n+m)$ amd $m \times (n+m)$, then:

$$\det\begin{pmatrix} X \\ Y \end{pmatrix} = \sum_{S\subseteq\{1,\dots,n+m\}, |S|=n} \operatorname{sign}(S) \cdot \det(X[S]) \det\left(Y[\overline{S}]\right)$$

where $\overline{S} = [n+m] - S$ and:

$$\operatorname{sign}(S) = (-1)^{\#\{\overline{s}<s \mid s\in S,\ \overline{s}\in\overline{S}\}}$$

Corollary 0.0.1. Let A, B be $n \times m$ and $m \times n$ matrcies respectively, with $n \leq m$, then:

$$\det\begin{pmatrix} 0_{n\times n} & A \\ B & \mathbb{I}_{m\times m} \end{pmatrix} = \sum_{|S|=n} (-1)^n \det(A[S]) \det(B^\intercal[S])$$

Proof of Cauchy-Binet Theorem 0.0.1. Recall that A, B has $n \times m$ and $m \times n$ dimension respectively. Notice that:

$$\begin{pmatrix} AB & 0_{m\times m} \\ B & \mathbb{I}_{m\times m} \end{pmatrix} = \begin{pmatrix} -\mathbb{I}_{n\times n} & A \\ 0_{m\times n} & \mathbb{I}_{m\times m} \end{pmatrix} = \begin{pmatrix} 0_{n\times m} & A \\ B & \mathbb{I}_{m\times m} \end{pmatrix}$$

Take determinant on both sides:

$$\begin{aligned}\det(AB) = (-1)^n \cdot 1 \cdot \det \begin{pmatrix} 0_{n\times n} & A \\ B & \mathbb{I}_{m\times m} \end{pmatrix} &= (-1)^{2n} \sum_{|S|=n} \det(A[S]) \det(B^\intercal[S]) \\ &= \sum_{|S|=n} \det(A[S]) \det(B^\intercal[S])\end{aligned}$$

■

0.1 Graph Eigenvalues

To better phrasing a graph, we first need to define multiset so that we can express not only simple graph but also general graph in a mathematically rigourous way.

Definition 0.1.1 (Multiset). A multiset M on a set S is an unordered collection of elements in S, s.t.

1. $\forall\, x \in M, x \in S$.
2. The # of times for $x \in S$ to appear in M, denoted as $\mu_M(x)$, is ≥ 0.

Example 0.1.1. If $\mu_M(x) = 0, 1\ \forall x$, then M is a set.

Note that two multiset M, M' are said to be equal if $\forall x \in S, \mu_M(x) = \mu_{M'}(x)$.

Notation. Let S be a finite set of size p, then define:

$$\binom{S}{k} = \{k - \text{subsets of } S\}$$

note that

$$\left|\binom{S}{k}\right| = \binom{|S|}{k}$$

also define:

$$\left(\!\binom{S}{k}\!\right) = \{k - \text{subsets of } S\}$$

note that:

$$\left|\left(\!\binom{S}{k}\!\right)\right| = \binom{p+k-1}{k}$$

this is the case: consider rephrasing the combinatorial problem as possible assigning of numbers for p numbers $a_1, \ldots, a_p$ with $a_1 + \ldots + a_p = k$ and $a_i \in [\![0, k]\!]$.

Thus we can define the graph properly.

Definition 0.1.2 (Graph). A finite graph is a triple $G = (V, E, \varphi)$ with:

- $V = \{v_1, \ldots, v_p\}$.
- $E = \{e_1, \ldots, e_q\}$.
- φ is a function $E \to \left(\!\binom{V}{2}\!\right)$.

A finite simple graph is the same data with $\varphi : E \to \binom{V}{2}$

Definition 0.1.3 (Adjacency Matrix of Graph). The adjacency matrix of a graph G, denoted as

$A(G)$, whose entries is defined by:

$$a_{ij} = \left|\varphi^{-1}(\{v_i, v_j\})\right|$$

In particular it counts the number of edges between two vertices v_i and v_j. Note that it is well-defined since if there is no edges between v_i and v_j, then the preimage of φ will be $\emptyset$, thus $a_{ij} = 0$.

Definition 0.1.4 (Walk). A walk of length k in a graph G is a non-empty finite sequence of vertices and edges

$$W = v_0, e_1, v_1, e_2, v_2, \ldots, e_k, v_k$$

such that for all $1 \leq i \leq k$, the edge e_i has **endpoints** v_{i-1} and v_i.

In a simple graph, where the edges are determined by their endpoints, a walk can be simplified to a sequence of vertices:

$$W = (v_0, v_1, \ldots, v_k) \quad \text{where } \{v_{i-1}, v_i\} \in E(G)$$

Note.

- In a walk, the both the edges and vertices can appear **repeatedly**.
- If $v_0 = v_k$, then such walk is called a **closed walk**.

Proposition 0.1.1. For any integer $l \geq 1$, the (i, j) entry of $(A(G))^l$, denoted as a_{ij}, is equal to the # of walks of length l in G starting from v_i to v_j.

Theorem 0.1.1. Let G be graph with $A(G)$ possessing eigenvalues $\lambda_1, \ldots, \lambda_p$, the # **closed walks** of length l is:

$$f_G(l) = \sum_{i=1}^{p} \lambda_i^l$$

The proof is straightforward combining the **Proposition** 0.1.1 and **Lemma** 0.0.2.

Definition 0.1.5 (Unitary Matrix). A complex square matrix U is unitary if $U^* = U^{-1}$.

Definition 0.1.6 (Orthogonal Matrices). An orthogonal matrix is a real square matrix O whose rows and cols are **orthonormal**, in particular equivalent to:

$$O^\intercal = O^{-1}$$

Theorem 0.1.2 (Spectral Theorem for Hermitian Matrices). A complex (real) square matrix A is Hermitian (real symmetric). Then there exists a unitary (orthogonal) U and a real diagonal matrix Λ, s.t.

$$A = U\Lambda U^{-1}$$

where

$$\Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$$

In particular, U can be chosen to have the orthonormal eigenvectors of A as its column vectors.

Corollary 0.1.1. There are $(p-1)^l + (-1)^l(p-1)$ closed walks in K_p.

Notation.

- $\mathbb{J}$ is the all 1 matrix.
- $\mathbb{I}$ is the identity matrix.

Proof. For K_p, the adjacency matrix is given by $A(K_p) = \mathbb{J} - \mathbb{I}$, with the eigenvalue of $\mathbb{J}$ being $\{p, 0, \ldots, 0\}$ and the eigenvalues of $\mathbb{I}$ being $\{1, \ldots, 1\}$. By **Lemma** 0.0.3, here we have $f(x) = x - 1$, with $\mathbb{J}$ pluggined, thus yields the result. ■

Corollary 0.1.2. There are $\frac{1}{p}((p-1)^l - (-1)^l)$ non-closed walks of length l in K_p.

Proof. Consider the matrix $A(K_p)^l = (\mathbb{J} - \mathbb{I})^l$, since $\mathbb{J}$ and $\mathbb{I}$ commutes, one can expand it using the binomial theorem:

$$(\mathbb{J} - \mathbb{I})^l = \sum_{i=0}^{l} (-1)^{l-i} \binom{l}{i} \mathbb{J}^i$$

And note that:

$$\mathbb{J}^i = \begin{cases} \mathbb{J}^0 = \mathbb{I} & i = 0 \\ p^{i-1}\mathbb{J} & i > 0 \end{cases}$$

Then see that:

$$\begin{aligned}
\sum_{i=0}^{l} (-1)^{l-i} \binom{l}{i} \mathbb{J}^i &= \left(\sum_{i=0}^{p} (-1)^{l-i} \binom{l}{i} p^{i-1} \right) \mathbb{J} + (-1)^l \mathbb{I} \\
&= \left(\sum_{i=0}^{p} (-1)^{l-i} \binom{l}{i} p^{i-1} - (-1)^l \frac{1}{p} \right) \mathbb{J} + (-1)^l \mathbb{I} \\
&= \left(\frac{1}{p}(p-1)^l - (-1)^l \frac{1}{p} \right) \mathbb{J} + \underbrace{(-1)^l \mathbb{I}}_{\text{contribute nothing}}
\end{aligned}$$

■

One can note that λ_i of $A(G)$ is completely determined by traces of $A(G)^l \quad \forall l \geq 1$ by **Lemma** 0.0.5. In particular if we know enough number of traces of $A(G)^l$ for multiple l, then we can calculate out the eigenvalues.

0.2 Radon Transform

We may define inner product space and orthogonal stuff first.

Definition 0.2.1 (Inner product space). An inner product space is a vector space over $\mathbb{C}$ together with an inner product $\langle -, - \rangle : V \times V \to \mathbb{C}$, s.t. $x, y, z \in V, \quad a, b \in \mathbb{C}$:

1. Conjugate symmetry: $\langle x, y \rangle = \overline{\langle y, x \rangle}$.
2. Linearity with the first entry: $\langle ax + by, z \rangle = a\langle x, z \rangle + b\langle y, z \rangle$.
3. Positivity: if $x \neq 0$, then $\langle x, x \rangle > 0$.

In this section we shall assume all V to be inner product space.

Definition 0.2.2. $x, y \in V$ are orthogonal if $\langle x, y \rangle = 0$.

Lemma 0.2.1. If $\langle x, y \rangle = 0$, then x, y are linearly independent.

Definition 0.2.3 (Kronecker Delta). Let S be a set, the kronecker delta function on S^2 is given by:

$$\delta_{uv} = \begin{cases} 1, & u = v \\ 0, & u \neq v \end{cases}$$

Let $\mathbb{Z}_2$ be the cyclic group of order 2, i.e. $(\{0,1\},+)$.
Let $\mathbb{Z}_2^n$ be the n-folde product of $\mathbb{Z}_2$, called an n-cube:

$$\mathbb{Z}_2^n = \{(a_1,\dots,a_n) \mid a_i \in \mathbb{Z}_2\}$$

such set has some properties if viewed as a vector space, for example, ZZ_2^n has a dot product defined by:

$$\langle -,- \rangle : \mathbb{Z}_2^n \times \mathbb{Z}_2^n \to \mathbb{Z}_2$$
$$\langle y,z \rangle \mapsto \underbrace{\sum y_i z_i}_{\text{group adding}} \in \mathbb{Z}_2$$

Lemma 0.2.2. $\forall u,v,w \in ZZ_2^n$, see that $u+v=w \Leftrightarrow u+w=v \Leftrightarrow v+w=u$.

The proof directly follows once realize that in this group $(-)$ is the same as $(+)$.
For $u \in \mathbb{Z}_2^n$, the weight $|u| = \sum u_i$ which count the number of entries that is non-zero.

0.2.1 Counting in Cube

In this section we shall use Radon Transform to help counting the closed walks in a cube. For other special graphs, one can similarly embed the structure of it into a specific group and construct the corresponding Radon Transform to count. It may rely on specific kinds of symmetry but really provide and powerful and convenient tool for counting.

Proposition 0.2.1. Let V be all functions $f : \mathbb{Z}_2^n \to \mathbb{C}$. This is a vector space over $\mathbb{C}$, we have the following fact:

1. $\dim_{\mathbb{C}} V = 2^n$, with the basis given by:

$$i_u(v) = \begin{cases} 1, & \text{if } u = v \\ 0, & \text{if } u \neq v \end{cases}$$

2. V has a inner product space structure over $\mathbb{C}$:

$$\langle f,g \rangle = \sum_{u \in \mathbb{Z}_2^n} f(y)\overline{g(u)}$$

3. V has basis:

$$\mathcal{B}_1 = \{f_u : \ u \in \mathbb{Z}_2^n,\ f_u(v) = \delta_{uv}\}$$
$$\mathcal{B}_2 = \{\chi_u : \ \chi_u(v) = (-1)^{uv}\}$$

See that:

$$g(v) = \sum_{u \in \mathbb{Z}_2^n} g(v)\delta_{uv} = \sum_{u \in \mathbb{Z}_2^n} g(v) f_u(v)$$

and

$$
\begin{aligned}
\langle \chi_u, \chi_v \rangle &= \sum_{w \in \mathbb{Z}_2^n} \chi_u(w)\overline{\chi_v(w)} \\
&= \sum_{w \in \mathbb{Z}_2^n} (-1)^{(u+v)\cdot w} \\
&= \begin{cases} 2^n, & \text{if } u = v \Leftrightarrow 2u = 0 \\ 0, & \text{if } u \neq v \end{cases}
\end{aligned} \tag{1}
$$

which means that χ_u and χ_v are orthogonal and thus linearly independent. The part that leads to 0 can be deduce by pairing vector w on the entry that $(u+v)$ is 1.

Definition 0.2.4 (Radon Transform). For a subset $\Gamma \in \mathbb{Z}_2^n$ and a function $f \in V : \mathbb{Z}_2^n \to \mathbb{C}$, the discrete radon transform on Γ of $\mathbb{Z}_2^n$ is:

$$\Phi_\Gamma(f)(v) := \sum_{w \in \Gamma} f(v+w)$$

this is a linear transformation $V \to V$.

Theorem 0.2.1. The eigenvectors of Φ_Γ are $\{\chi_u\}_{u \in \mathbb{Z}_2^n}$, with eigenvalues being $\lambda_u = \sum_{w \in \Gamma} (-1)^{u \cdot v}$.

Proof. See that:

$$
\begin{aligned}
\Phi_\Gamma(\chi_u)(v) &= \sum_{w \in \Gamma} \chi_u(v+w) \\
&= \sum_{w \in \Gamma} (-1)^{u(v+w)} \\
&= (-1)^{uv} \sum_{w \in \Gamma} (-1)^{uw} \\
&= \chi_u(v) \cdot \lambda_u
\end{aligned}
$$

■

Definition 0.2.5. The n-cube graph C_n is the graph with:

- $V(G) = \mathbb{Z}_2^n$.
- $E(G) = \{(u,v) \mid u, v \text{ differ in only 1 coordinate}\}$, for example the 3-dimensional cube.

The adjacency matrix of this kinds of graph is rather complicated and hard to comprehend and compute its eigenvalue by simply looking at the matrix.

We now propose a simpler way to obtain the eigenvalue and eigenvector of adjacency matrix of n-cube graph, basically choosing special subset to do radon transform and see that the matrix is exactly the same.

Let Δ be the set:

$$\Delta := \{\delta_i \mid \delta_i \text{ is the i-th unit vector in } \mathbb{Z}_2^n\} \subseteq \mathbb{Z}_2^n$$

Let $\Phi_\Delta : V \to V$, and $[\Phi_\Delta]$ be the matrix of Φ_Δ w.r.t. $\mathcal{B}_1 = \{f_u : \ u \in \mathbb{Z}_2^n\}$.

Lemma 0.2.3. The matrix $[\Phi_\Delta] = A(C_n)$.

Proof. Basically we want to see that all entries on the left hand side is the same as the right hand side, so we consider the (u,v)-entry of $[\Phi_\Delta]$, let $z \in \mathbb{Z}_2^n$, and compute: $\Phi_\Delta f_u(v)$,

$$\begin{aligned}
\Phi_\Delta f_u(z) &= \sum_{w\in\Delta} f_u(z+w) \\
&= \sum_{w\in\Delta} f_{u+w}(z) \\
\Rightarrow \Phi_\Delta f_u &= \sum_{u\in\Delta} f_{u+w} \\
\Rightarrow [\Phi_\Delta]_{u,v} &= \begin{cases} 1, & \text{if } u+w=v \\ 0, & \text{if } u+w\neq v \end{cases} \\
\Leftrightarrow [\Phi_\Delta]_{u,v} &= \begin{cases} 1, & \text{if } u+v=w\in\Delta \\ 0, & \text{otherwise} \end{cases} \\
\Leftrightarrow [\Phi_\Delta]_{u,v} &= \begin{cases} 1, & \text{if } u,v\in E(C_n) \\ 0, & \text{otherwise} \end{cases}
\end{aligned}$$

■

Corollary 0.2.1. The eigenvectors of $A(C_n)$ are:

$$\mathcal{E}_u = \sum_{v\in\mathbb{Z}_2^n} (-1)^{u\cdot v} v$$

With eigenvalues being:

$$\lambda_u = n - 2|u|$$

In particular, $A(C_n)$ has $\binom{n}{k}$ eigenvalues equal to $n-2k$.

Proof. For $g \in V$, see that $g = \sum_{v\in\mathbb{Z}_2^n} g(v)\cdot f_v$, which implies that:

$$g(u) = \sum_{v\in\mathbb{Z}_2^n} g(v) f_v(u) = \sum_{v\in\mathbb{Z}_2^n} g(v)\delta_{uv}$$

Recall that $[\Phi_\Delta]_{\mathcal{B}_1}$ has eigenvectors $\{\chi_u(v) = (-1)^{u\cdot v}\}$, then:

$$\chi_u = \sum_{v\in\mathbb{Z}_2^n} \chi_u(v) f_v = \sum_{v\in\mathbb{Z}_2^n} (-1)^{u\cdot v} f_v$$

with eigenvalues being:

$$\lambda_u = \sum_{w\in\Delta} (-1)^{u\cdot w} = \sum_{i\in[n]} (-1)^{u\cdot\delta_i} = n - 2|u|$$

Notice that those $\mathcal{E}_u$ with the **Hamming Weight** of u being the same share the same eigenvalue, total number of such u will be $\binom{n}{k}$ if $|u| = k$. ■

Proposition 0.2.2. If G is a graph with p vertices and eigenvalues $\lambda_1, \dots, \lambda_p$, then there exists $c_1, \dots, c_p \in \mathbb{R}$, s.t.

$$A(G)^l_{ij} = \sum_{k=1}^{p} c_k \lambda_k^l$$

And in particular, if U is an **orthogonal** matrix ($A(G) = U\Lambda U^{-1}$), then:

$$A(G)^l_{ij} = \sum_{k=1}^{n} U_{ik} U_{jk} \lambda_k^l$$

Proof. By **Spectral Theorem** 0.1.2, there exists an orthogonal matrix U, s.t.

$$A = U\Lambda U^{-1}$$

Let $l \geq 1$, then:

$$A^l = U\Lambda^l U^{-1}$$

take the (i,j)-entry of right hand side and see that it equal to $\sum_{k=1}^{n} U_{ik}U_{jk}\lambda_k^l$. ■

Corollary 0.2.2. Let $u, v \in \mathbb{Z}_2^n$, supppose that $|u+v| = k$, i.e. u and v **differ** in k places, then:

$$A(C_n)_{uv}^l = \frac{1}{2^n}\sum_{i=1}^{n}\sum_{j=1}^{k}(-1)^k\binom{k}{j}\binom{n-k}{i-j}(n-2i)^l \qquad (i \geq j)$$

Proof. Consider the eigenvector of $A(C_n)$, given by:

$$\mathcal{E}_u = \sum_{v\in\mathbb{Z}_2^n}(-1)^{u\cdot v}v$$

By **Equation** 1, $|\mathcal{E}_u| = 2^{\frac{n}{2}}$, so one needs to normalize it to get the orthonormal basis:

$$\mathcal{E}'_u = \frac{1}{2^{\frac{n}{2}}}\mathcal{E}_u$$

Then by **Proposition** 0.2.2, we have:

$$A(C_n)_{uv}^l = \frac{1}{2^n}\sum_{w\in\mathbb{Z}_2^n}\mathcal{E}'_{uw}\mathcal{E}'_{vw}\lambda_w^l$$

One can then consider what is $\mathcal{E}'_{uw}$, written in canonical basis $v \in \mathbb{Z}_2^n$, it is exactly:

$$\mathcal{E}'_{uw} = (-1)^{u\cdot w}$$

Thus we have:

$$A(C_n)_{uv}^l = \frac{1}{2^n}\sum_{w\in\mathbb{Z}_2^n}(-1)^{(u+v)\cdot w}(n-2|w|)^l$$

One may consider the red part, with the number of vectors w of **Hamming Weight** i which have j 1's in common with $u+v$ is:

$$\binom{k}{j}\binom{n-k}{i-j}$$

it is simply the way of how we choose w, so that the dot product result $(\mathbf{u}+\mathbf{v})\cdot\mathbf{w} = \mathbf{j}$. As w runs over all elements in $\mathbb{Z}_2^n$, where k is just $|u+v|$ to be the **upper bound** of j, see that:

$$A(C_n)_{uv}^l = \frac{1}{2^n}\sum_{i=1}^{n}\sum_{j=1}^{k}(-1)^j\binom{k}{j}\binom{n-k}{i-j}(n-2i)^l \quad (i = |w|)$$

■

0.3 Matrix-Tree Theorem

Definition 0.3.1 (Spanning Subgraph). A spanning subgraph of a graph G is a graph H with the same vertex set as G and $E(H) \subseteq E(G)$.

(If G has q edges, there are 2^q spanning subgraph, in particular, choose whether each edges are included in.)

Definition 0.3.2 (Spanning Tree). A spanning tree is a spanning subgraph that is a **tree**. The # of spanning tress in a graph G is called complexity of G, denoted as $\kappa(G)$.

Definition 0.3.3 (Laplacian of a Graph). The laplacian of a graph G is given by:

$$\mathbb{L}(G) = \begin{cases} -\left|\varphi^{-1}(v_i, v_j)\right|, & \text{if } v_i \neq v_j \\ \deg(v_i), & \text{otherwise} \end{cases}$$

In short, the diagonal of such matrix will be the degree of the vertex, and the rest represent the number of edges connect between the vertices mult with -1.

Note. The laplacian actually can also be given by:

$$\mathbb{L}(G) = \mathrm{diag}\{\deg(v_1), \dots, \deg(v_n)\} - A(G)$$

Definition 0.3.4 ((r,c)-Cofactor). Given a matrix A, the (r,c)-cofactor of A is the:

$$A_{r,c} := (-1)^{r+c} \det A_{\hat{r},\hat{c}}$$

where $\hat{r}$ represent deleting the r-th row, and $\hat{c}$ represent deleting the c-th column.

Definition 0.3.5 (Orientation). An orientation of a graph G is an assignment of an order on every edge: which is, for every edge $e \in E(G)$, for example, it connects u and v, then choose one of the ordered pair (u, v) or (v, u) as direction. Such orientation is usually denoted as $\mathfrak{o}$.

Definition 0.3.6 (Incidence Matrix). The incident matrix of a directed graph G with respect to the orientation $\mathfrak{o}$ is the $|V| \times |E|$ matrix whose (i, j)-entry is given by:

$$\mathbb{M}(G) = \begin{cases} -1, & \text{if the edge } e_j \text{ has initial vertex } v_i \\ 1, & \text{if the edge } e_j \text{ has final vertex } v_i \\ 0, & \text{otherwise} \end{cases}$$

Lemma 0.3.1.

$$\mathbb{L}(G) = \mathbb{M}(G)\mathbb{M}(G)^\intercal$$

Proof. The proof is quite straightforward once consider the situation of matrix manipulation, first denote $\mathbb{M}(G) = (m_{ij})$:

$$(\mathbb{M}(G)\mathbb{M}(G)^\intercal)_{ij} = \sum_{e_k \in E(G)} m_{ik} m_{jk}$$

Consider the situation of $i = j$ and $i \neq j$:

- When $i \neq j$: One should only consider the case when e_k connects i and j, otherwise one of them will be 0 and thus the product of them will be 0. So consider the case where i and j are connected by e_k. In this case one vertex must be the source and one vertex must be the sink, thus one of them will be 1 and the other will be -1. Summing through all edges in the graph gives us the result to be $-\left|\varphi^{-1}(v_i, v_j)\right|$.
- When $i = j$: In this case one will still only consider the case where e_k touches $v_i = v_j$, otherwise one of m_{ik} or m_{jk} will be 0. Whether it is the source or sink it doesn't matter, the result should still yields to be 1. Summing through all edges in the graph gives us $\deg(v_i = v_j)$.

It follows that this is exactly the **Definition** 0.3.3 of Laplacian. ■

Lemma 0.3.2. Let $|V| = p$. Let S be a subset of $p-1$ edges. Let v be any vertex of G. Then:

$$\det\left(\underbrace{\mathbb{M}_{\hat{v}}[S]}_{(p-1)\times(p-1)}\right) = \begin{cases} \pm 1, & \text{if } S \text{ forms a spanning tree of } G \\ 0, & \text{otherwise} \end{cases}$$

Proof. One can prove the theorem by deviding the case of whether S forms a spanning tree of G:

- If S doesn't forms a spanning tree. Then there must exsist a cycle C in the graph, denoted the edge of the cycle as $f_1, \ldots, f_s$. The point is we want to make it as a oriented cycle, so that the column vector representing $f_1, \ldots, f_s$ are linearly dependent. For convenience one can interchange the column vector of $f_1, \ldots, f_s$ into in order and adacent ones on the matrix, the determinant still remains the same. One can choose an directed orientation on such edges set, for example in the sense that $1 \to 2 \to \cdots \to s$, and for those edges with the same orientation with the chosen one, we multiply the column vectors with 1 and for those who are not, we multiply with -1 for the column vectors. One can see that by definition, summing over all such column vectors with the result leads to 0, thus $f_1, \ldots, f_s$ are linearly dependent, thus the determinant will be 0.

- If S did forms a spanning tree. By degree handshake theorem, one can see that the tree always have a leave vertex, namely a vertex that only connects with a unique edge. We now choose a leave node v in the spanning tree and the corresponding edge e, by definition of incidence matrix, for the row represented by v attains only (v, e)-entry which is non-zero, and it will be ± 1. By property of determinant, abuse the notation of S for the whole spanning tree, see that $\det(S) = \det(S \backslash \{v\})$. This is true as we also deleted the corresponding edges as we deleted the vertex. And we induct on the leavenode to reduct on the whole graph, yields the result to be ± 1.

■

Theorem 0.3.1 (Matrix-Tree Theorem). Let G be a finite connected graph without loops. The # of spanning trees of G is:

$$\kappa(G) = L_{r,c}(G) \qquad \forall\, r, c$$

where $L_{r,c}$ is the (r, c)-cofactor of $\mathbb{L}(G)$. Note that the following prove only completes for (v, v)-cofactor, and combined with **Lemma** 0.3.3 the prove completes.

Proof. By the fact that $\mathbb{L} = \mathbb{M}\mathbb{M}^\intercal$, we have:

$$\mathbb{L}_{\hat{v}} = \mathbb{M}_{\hat{v}}\mathbb{M}_{\hat{v}}^\intercal$$

Now apply **Cauchy-Binet Formula** 0.0.1:

$$\det(\mathbb{L}_{\hat{v}}) = \sum_{|S|=p-1, S\subseteq[|E|]} \det(\mathbb{M}_{\hat{v}}[S])\det(\mathbb{M}_{\hat{v}}[S]^\intercal) \xlongequal{\text{Lemma } 0.3.2} \kappa(G)$$

■

Lemma 0.3.3. If $\mathbb{A} = (a_{ij})$ is a square matrix of $n \times n$ dimension, s.t.

- all the row sums are zero.
- all the column sums are zero.

then:

1. for $1 \le r \le n$, have:
$$A_{r,1} = A_{r,2} = \cdots = A_{r,n}$$
2. for $1 \le c \le r$
$$A_{1,c} = A_{2,c} = \cdots = A_{n,c}$$

Namely all its cofactors are the same.

Proof. Fix r, let $\mathbb{A}'$ be the matrix that has $a'_{r,1} = a_{r,1} + 1$, which bump the row sum of row r to be 1 instead of 0, and the other entries remains the same as $\mathbb{A}$. Choose a $k \in [\![1, n]\!]$. Let $\mathbb{A}''$ be the matrix with the k-th column equal to the sum of all columns of $\mathbb{A}'$, and the other columns the same as $\mathbb{A}'$. As the column sum will be 0 except for row r, the k-column of $\mathbb{A}''$ will admit 1 at (r, k) and 0 at the rest of the entries. We have following observation:

- $\det \mathbb{A}' = \det \mathbb{A}''$. Since $\mathbb{A}''$ is obtained by doing elementary operation from $\mathbb{A}'$.
- $\det \mathbb{A}'' = A''_{r,k}$ by the fact that the k-th column only admits 1 at r-th row and 0 elsewhere.
- $\det \mathbb{A} + A_{r,1} = \det \mathbb{A}'$ by definition of determinant, and notice that $\det \mathbb{A} = 0 \Rightarrow A_{r,1} = A''_{r,k}$, and by construction of $\mathbb{A}''$ see that $A''_{r,k} = A_{r,k}$.

becase $\mathbb{A}'$ is independent of k, $A_{r,k}$ is independent of k. ■

Definition 0.3.7 (d**-Regular Graph**)**.** A graph is d-regular if all vertices have degree to be d.

Proposition 0.3.1. If G is d-regular on n vertices, the $\mathbb{L}(G) = d \cdot \mathbb{I}_n - \mathbb{A}(G)$. In particular, if G has eigenvalues to be $\lambda_1, \ldots, \lambda_n$ with $\lambda_n = d$, then $\mathbb{L}(G)$ has eigenvalues $d - \lambda_1, \ldots, d - \lambda_{n-1}, d - \lambda_n = 0$. Moreover:
$$\kappa(G) = \frac{1}{n} \prod_{i=1}^{n-1} (d - \lambda_i)$$

Proof. Since $\mathbb{L}(G)$ has zero row and column sums, cofactors of $\mathbb{L}(G)$ are equal by **Lemma** 0.3.3. Compute the characteristic polynomial of $\mathbb{L}(G)$:
$$\begin{aligned}
\det(\mathbb{L}(G) - t \cdot \mathbb{I}_n) &= \det(\mathbb{L}(G)) - t(L_1 + \cdots + L_{n,n}) + \text{higher degree terms} \quad \text{(results in linear algebra)} \\
&= \det(\mathbb{L}(G)) - ntL_{1,1} + \text{higher degree terms} \\
&= -ntL_{1,1} + \text{higher degree terms}
\end{aligned}$$

On the other hand,
$$\det(\mathbb{L}(G) - t\mathbb{I}_n) = \prod_{i=1}^{n} (\underbrace{\lambda'_i}_{\text{e.v. of } \mathbb{L}(G)} - t) = -t \underbrace{(d - \lambda_1) \cdots (d - \lambda_{n-1})}_{\text{eigenvalues of } \mathbb{L}(G) = d \cdot \mathbb{I}_n - \mathbb{A}(G)} + \text{higher degree terms}$$

thus
$$\kappa(G) = L_{i,i} = \frac{1}{n} \prod_{i=1}^{n-1} (d - \lambda_i)$$

■

Corollary 0.3.1. Let G be a connected loopless graph with p vertices. Suppose that the eigenvalues of $\mathbb{L}(G)$ are given by $\lambda_1, \ldots, \lambda_{n-1}, \lambda_n = 0$, then:
$$\kappa(G) = \frac{1}{n} \prod_{i=1}^{n-1} \lambda_i$$

Example 0.3.1.

1. Consider C_d be the cube graph with d dimensions, total vertices to be 2^d, have:
   - $\mathbb{A}(C_d)$ has $d - 2i$ as eigenvalues with multiplicity to be $\binom{d}{i}$ for $i \in [\![1, d]\!]$.
   - $\mathbb{L}(C_d)$ has eigenvalues $2i$ with multiplicity being $\binom{d}{i}$ for $i \in [\![1, d]\!]$.
   - Thus:
   $$\kappa(C_d) = \frac{1}{2^d} \prod_{i=1}^{d} (2i)^{\binom{d}{i}}$$
2. Consider K_p be the complete graph with p vertices.
   - $\mathbb{A}(K_p)$ has eigenvalues being $p - 1, -1, \dots, -1$.
   - $\mathbb{L}(K_p)$ has eigenavalues being $0, p, \dots, p$.
   - Thus:
   $$\kappa(K_p) = \frac{1}{p} \cdot (p^{p-1}) = p^{p-2}$$

0.4 Eulerian Tours

Definition 0.4.1 (Binary de Bruijn). Given a positive integer k, a binary de Bruijn sequence of degree k is a sequence of $0, 1$ that is periodic of period 2^k and that has the list of all contiguous subsequence to be exactly all binary strings of length k.

Namely we scan through the string, and any kinds of sequence consisting that alphabet with length k should only appear once.

Example 0.4.1.

- When $k = 2$, the following forms a binary de Bruijn sequence of degree 2.
$$1, 1, 0, 0, 1$$
- When $k = 3$, the following forms a binary de Bruijn sequence of degree 3.
$$0, 0, 0, 1, 1, 1, 0, 1, 0, 0, 0$$
which attains 000, 001, 011, 111, 101, 010, 100.

Definition 0.4.2 (Directed graph). A directed graph or a digraph D is (V, E, φ) where $\varphi : E \to V \times V$ which maps to ordered pair of vertices.

Definition 0.4.3 (Eulerian Tour). An Eulerian tour in G is a closed walk that visit all vertices and very edge exactly 1 times. Note that vertices can be visited many times.

Definition 0.4.4 (Arborescence). an arborescence is a directed graph where there exists a vertex r (called the root) such that, for any other vertex v, there is exactly one directed walk from r to v (noting that the root r is unique). If the underly tree is spanning in G_D, then we say the arborescence is spanning.

Definition 0.4.5.

- $\mathcal{E}(D, e) = \#$ of Eulerian tours that has its first edge to be e.
- $\tau(D, v) = \#$ of (spanning) arborescences rooted at v.

Definition 0.4.6 (Weakly Connected). A digraph D is weakly connected if D is connected as an undirected graph.

Definition 0.4.7 (Strongly Connected). A digraph D is strongly connected if there is a directed walk between any two vertices. Note that a walk goes from one of them to the other is enough.

Note. Clearly if a graph D is strongly connected, then it is weakly connected.

Lemma 0.4.1. Let D be a finite directed graph without isolated vertices, **TFAE**:

1. D is Eulerian.
2. D is connected as an undirected graph, and is balanced. (weakly connected + balanced)
3. D is connected directedly and balanced. (strongly connected + balanced)

A digraph is balanced if for any vertices v in D, the indegree of v is equal to the outdegree of v is equal to the outdegree of v.

Proof. Suppose that D is eulerian and there exists an eulerian tour $e_1, \ldots, e_q$, with the source being v. As we move along the tour, whenever we enter a vertex we must then exit it, except the final vertex of e_f without exiting it. But this is the source vertex so we exit it at the beginning, thus every vertex entered as often as exit, and they must have the same indegree and outdegree. Therefore D is balanced. Note that D is clearly strongly connected and thus weakly connected.

Conversely given a balanced, weakly connected digraph, we want to see that it is eulerian. The generalhidea is to first prove that for any vertex v in the digraph, there exists a tour starts with v and then ends with v. And then since D is a finite digraph, one can then choose a tour with maximum length r, and try to prove that $r = q$ be proceeding contradiction that if $r < q$, such will contradict to the maximality of r.

Since D is connected one may assume that D contains at least one edge e, take the source of that edge and denote it as v. Consider $e = (v, u)$, if $u = v$ and we are done. If $u \neq v$, since the indegree of u equals to the outdegree, then there is some exit edge e_1, either the target of e_1 being v and we are done, or being some vertex w, and continuing this way, either we complete a tour or else we have entered the current vertex once more than we have exited, which guaranteed by balanced that there exists an exit along a new edge other than previously in the tour. Since D has finitely many edges, evetually we must complete a tour. Thus D does have a tour that starts at any vertex v, and thus for any edge e, there exists a tour that use e.

One can then take the maximum size of tour, denoted as $e_1, \ldots, e_r$, one shall then want to prove that $r = q$. Suppose that $r < q$, then the edges form a subgrpah C of D. By definition, C should be also a balanced graph. Thus consider the graph $H := D \backslash C$ by removing all edges of C out of D, since D, C are both balanced graph, H shall also be a balanced graph, although may not connected. One shall then observe that there must exists a connected component H_1 in H that contains at least one edge, and a vertex v that is also contained in C, otherwise see that the tour is then directly being the eulerian tour and we have $r = q$, a contradiction. Thus we have such H_1 exists. But then see that in H_1 there exists a tour C' starts from v and ends with v by our previous reasoning, this gives us a "detour" as one can enlarge the tour by when moving along C, when arrive v, one can make detour on C' and back on v then proceed. This gives us a longer tour, contradicting to the maximality of r $\lightning$. Thus $r = q$ and yields the existence of Eulerian tour, and thus D is Eulerian. ∎

Theorem 0.4.1 (BEST Theorem — de Bruijn, van Aardenne-Ehrenfest, Smith, Tutte Theorem).

Let D be a connected balanced digraph with vertex set being V. Fix an edge e of D and let $v = init(e)$. Then
$$\mathcal{E}(D,e) = \tau(D,v)\prod_{u\in V}(\text{outdeg}_G(u)-1)!$$

Proof. Fix a vertex v and an edge e leaving v. For each Eulerian tour $\mathcal{E}$ that starts with v and e and each vertex $u \neq v$, consider the edge of first arrival at u.

Claim. The edges of first arrivals with e forms an arborescence T at v.

Let us order all incoming edge at every $u \neq v$ in the order of traversal in $\mathcal{E}$, these order have following property:

- Among all the incoming edges at $u \neq v$, the first edge belongs to T.
- At v, the edge e is the last in the ordering.

Lemma 0.4.2. There is a bijection between:
$$\{\text{Eulerian tours ending with } e \text{ at } v\}$$
$$\updownarrow$$
$$\left\{\begin{matrix}\text{Arborescence } T \text{ at } v, \text{ and orderings on} \\ \text{incoming edges satisfying the above conditions}\end{matrix}\right\}$$

Proof. Suppose we have

- Spanning arb at v.
- An ordering on incoming edges satisfying the above two bullet points.

Start with v and traverse e which the edge that enters v last. Traverse the graph by visiting the larger edge at each stop $u \neq v$. The only time we get stuck is when we visit v the last time (balanced).

Claim. At this point, we have traversed all edges in D.

Proof of Claim. Suppose otherwise, we left some edges out. Among the vertices incident to these edges, choose u to be the closest to v, otherwise have $u \neq v$. Since D is balanced, we have the same number of incoming edges and out comming edges at u, there is at least one incoming edge at u not traversed yet. By condition 1, we have not traversed the unique edge in T at u. Let u' be the vertex adjacent to e', with $u' \xrightarrow{e'} u$ by definition of u', u' must be strictly closer to v that u, leading to a contradiction $\lightning$. ■

■

■

Definition 0.4.8 (Directed Laplacian). Let D be a digraph with vertex set $V = \{v_1, \dots, v_p\}$ and with l_i loops at vertex v_i. Let $\mathbb{L}(D)$ be the directed laplacian defined by:
$$L_{ij} = \begin{cases} -m_{ij}, & \text{if } i \neq j \text{ and there are } m_{ij} \text{ edges with} \\ & \text{initial vertex } v_i \text{ and final vertex } v_j \\ \text{outdeg}(v_i) - l_i, & \text{if } i = j. \end{cases}$$

Theorem 0.4.2. $\det(\mathbb{L}_0) = \tau(D, v_p)$

0.5 The Sperner Property

Definition 0.5.1 (Partially Ordered Set). A poset P or partially ordered set is a finite set P together with a binary relation $\leq$ such that for all $x, y, z \in P$:

1. (reflexitivity): $x \leq x \; \forall \; x \in P$.
2. (anti-symmetry): If $x \leq y, y \leq x$ then $x = y$.
3. (transtivity): If $x \leq y, y \leq zs \Rightarrow x \leq z$.

Definition 0.5.2 (Cover). If $x < y$ and no $z \in P$ satisfies $x < z < y$, then we say y covers x and write $x \lessdot y$.

Definition 0.5.3 (Hasse Diagram). The Hasse diagram of a poset P is the graph whose vertices are the elements of P and whose edges are the **cover relations**. We draw the Hasse diagram so that if $x < y$ then x is below y.

Definition 0.5.4 (Isomorphism between Posets). Two posets P and Q are isomorphic if there is a bijection $f : P \to Q$ such that for all $x, y \in P$, $x \leq y$ if and only if $f(x) \leq f(y)$.

Definition 0.5.5 (Chain). A chain C in a poset P is a totally ordered subset of P; that is, for all $x, y \in C$, either $x < y$ or $y < x$. A finite chain C has length n if C has $n + 1$ elements.

Definition 0.5.6 (Graded Poset). A finite poset P is graded of rank n if every maximal chain has length n. A maximal chain if it's contained in no larger chain.

Definition 0.5.7. If P is graded poset of rank n and $x \in P$, then there exists a rank function $\text{rk} : P \to \mathbb{N}$, s.t. x has rank j, denoted by $\text{rk}(x) = j$ if any maximal saturated chain of P with top element x has length j only welldefined when the poset is **graded**.

Definition 0.5.8. $P_j = \{x \in P \mid \text{rk}(x) = j\}$ and see that

$$P = \bigsqcup_{j=0}^{n} P_j$$

We are interested in: If a graded poset P of rank n has p_i elements of rank i, then we have a generating functions:

$$F(P, t) = \sum p_i t^i$$

Example 0.5.1. For boolean algebra $\mathbb{B}_n$, see that $F(P, t) = (1 + t)^n$ by binomial theorem.

Definition 0.5.9 (Rank Symmetric). A graded poset P of ran n is rank-symmetric if $p_0 = p_n$, $p_1 = p_{n-1}, \ldots$ etc.

Example 0.5.2. $\mathbb{B}_n$ is rank-symmetric since:

$$\binom{n}{0} = \binom{n}{n}, \quad \binom{n}{1} = \binom{n}{n-1}, \quad \cdots$$

Definition 0.5.10 (Unimodal). A graded poset P of rank n is unimodal if there exists some k such that

$$p_0 \leq p_1 \leq \cdots \leq p_{i-1} \leq p_i \geq p_{i+1} \geq \cdots \geq p_n$$

Definition 0.5.11 (Antichain). An anti-chain in P is a subset of P where **no two** elements are comparable.

Theorem 0.5.1 (Sperner's Theorem). The largest collection of subsets of $[n]$ such that no two contain each other is

$$\binom{[n]}{\lfloor n/2 \rfloor} = \text{ the subsets of } [n] \text{ of size } \lfloor n/2 \rfloor$$

Definition 0.5.12 (Sperner's Property). The poset P has Sperner's property if the size of the largest antichain in P is equal to the $\max\{P_i \mid 0 \leq i \leq n\}$

Definition 0.5.13 (Order Matching). An order matching from P_i to P_{i+1} is an injective function:

$$\mu : P_i \to P_{i+1}$$

satisfying that $x \lessdot \mu(x)$.

Appendix